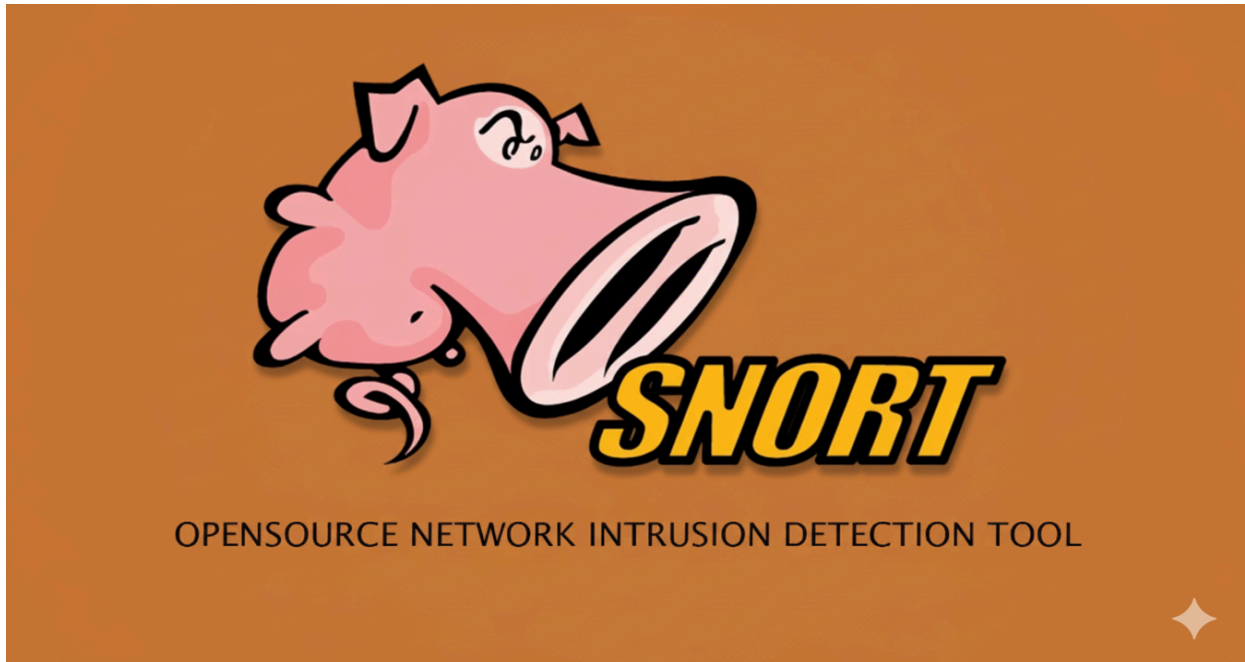


SNORT TOOL



1.What is Snort?

Snort is an **open-source Intrusion Detection and Prevention System (IDS/IPS)** developed by **Cisco**.

It helps you **detect and block malicious network traffic** — like hacking attempts, malware communication, or suspicious activities.

Example:

Snort is like a **security camera for your network** 📹

It **captures everything, recognizes bad behavior**, and can **sound an alarm** or even **lock the door** to stop the attacker.

2.Why Do We Use Snort?

We use **Snort** to **protect networks** from malicious activities by **monitoring, detecting, and preventing** attacks in real time.

In short — **Snort helps you see what's happening inside your network and react when something suspicious occurs.**

3. Why Snort is Popular

- **Free and open-source** — anyone can use and customize it.
- **Lightweight and flexible** — works on small networks or large enterprises.
- **Community-supported** — regular updates from Cisco Talos and open-source contributors.
- **Educational** — great for learning how network attacks actually look in packet data.
- **Integrates well** — can connect with SIEM tools (like Splunk, Wazuh, or ELK).

4. How Snort Works

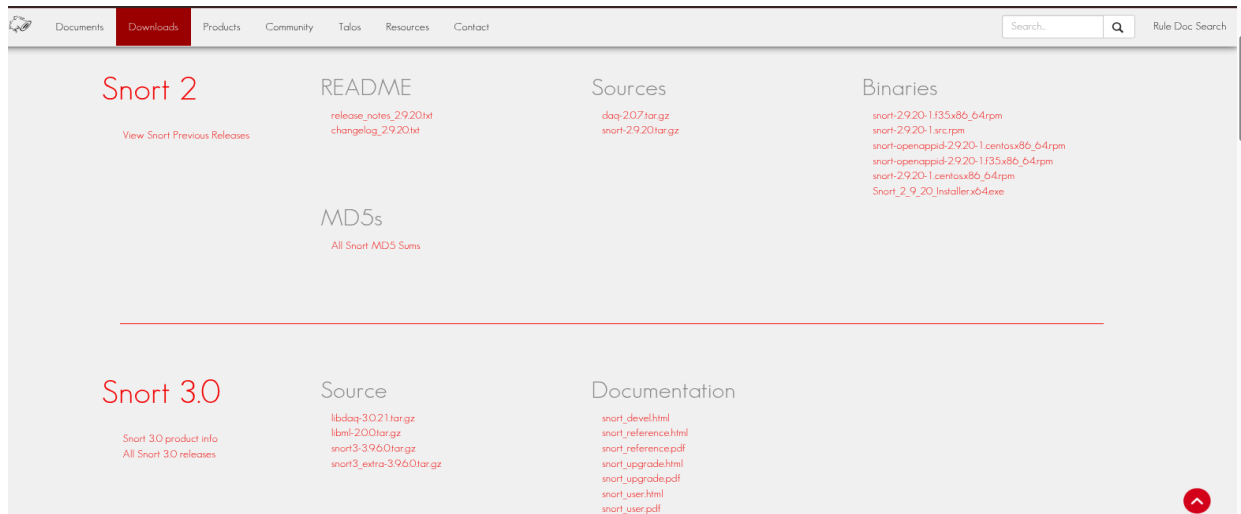
Think of Snort as a **security guard** at your network's gate 🙋

1. **Packet Decoder** — reads raw packets coming from the network.
2. **Preprocessors** — normalizes traffic and prepares it for analysis (e.g., defragmenting).
3. **Detection Engine** — compares packets to Snort **rules**.
4. **Output Module** — logs or alerts you if something suspicious is found.

5.How to install snort:-

For windows:

<https://www.snort.org/>



For-linux :

sudo apt update

sudo apt install snort -y

```
(yuvi@kali)-[~]
└─$ snort -h
Snort has several options to get more help:

-? list command line options (same as --help)
--help this overview of help
--help-commands [<module prefix>] output matching commands Help
--help-config [<module prefix>] output matching config options
--help-counts [<module prefix>] output matching peg counts
--help-limits print the int upper bounds denoted by max*
--help-module <module> output description of given module
--help-modules list all available modules with brief help
--help-modules-json dump description of all available modules in JSON format
--help-plugins list all available plugins with brief help
--help-options [<option prefix>] output matching command line options
--help-signals dump available control signals
--list-buffers output available inspection buffers
--list-builtin [<module prefix>] output matching builtin rules
--list-gids [<module prefix>] output matching generators
--list-modules [<module type>] list all known modules
--list-plugins list all known modules
--show-plugins list module and plugin versions

--help* and --list* options preempt other processing so should be last on the
command line since any following options are ignored. To ensure options like
--markup and --plugin-path take effect, place them ahead of the help or list
options.

Options that filter output based on a matching prefix, such as --help-config
won't output anything if there is no match. If no prefix is given, everything
matches.

Report bugs to bugs@snort.org
```

6.Understanding IDS, IPS & Snort's Role

Intrusion Detection System (IDS)

- **IDS** is a **passive monitoring system** that focuses on **detecting** suspicious activities or policy violations.
- It **doesn't block** traffic — it only **alerts** when something abnormal is detected.

Two Main Types of IDS:

Type	Description	Example
Network Intrusion Detection System (NIDS)	Monitors network traffic across multiple systems or subnets.	Detects if someone is scanning all devices in your office network.
Host-based Intrusion Detection System (HIDS)	Monitors activities on a single device or endpoint.	Detects if a user tries to modify system files or registry entries.

Intrusion Prevention System (IPS)

- **IPS** is an **active protection system** that not only detects threats but also **prevents or blocks** them in real time.
- It acts as a **security guard** who both **notifies** and **stops** suspicious activity.

Four Main Types of IPS:

Type	Description	Example
Network Intrusion Prevention System (NIPS)	Monitors and protects entire network subnets.	Drops packets from a known malicious IP.
Behaviour-based IPS (NBA)	Detects abnormal behavior by learning what “normal” traffic looks like.need training time period to learn(baselining)	Identifies new types of malware or DDoS patterns.
Wireless Intrusion Prevention System (WIPS)	Protects wireless networks from attacks.	Detects rogue Wi-Fi access points or unauthorized connections.
Host-based Intrusion Prevention System (HIPS)	Protects a single device by blocking malicious activity locally.	Prevents malware from modifying system processes.

Detection and Prevention Techniques

Both IDS and IPS use **three key techniques** to detect or prevent attacks:

Technique	Approach	Example
Signature-Based	Uses known patterns or signatures to detect threats.	Detects a virus known by its signature.
Behaviour-Based	Learns normal behavior and flags anything abnormal.	Detects a new, previously unknown malware.
Policy-Based	Detects when system activity violates security policies.	Alerts if users access restricted files.

7.Capabilities of Snort

- Live network traffic analysis
- Attack and probe detection
- Packet logging and inspection
- Protocol analysis
- Real-time alerting
- Modular and plugin-based
- Supports preprocessors (for cleaning and preparing data)
- Works on both Linux and Windows

8.Snort Operating Modes

Mode	Description	Example
Sniffer Mode	Reads IP packets and displays them in the console.	Monitor live network packets in real-time.
Packet Logger Mode	Logs all IP packets (incoming/outgoing) for later review.	Analyze captured traffic after an attack.
NIDS/NIPS Mode	Detects or blocks malicious packets using rules.	Detect and prevent port scans or brute-force attacks.

First Interaction with Snort

1. Check Snort Version

```
snort -V  or  snort - -version
```

2. Verify the Configuration File

```
sudo snort -c /etc/snort/snort.conf -T
```

Parameter	Description
-c	Specifies the configuration file to use
-T	Runs a self-test to validate the setup
-V	Shows Snort version
-q	Quiet mode — prevents the default banner from displaying

9.Operation Mode 1: Sniffer Mode

Sniffer Mode Parameters

Parameter	Description
-v	Verbose — shows packet info like TCP/IP headers (tcpdump-like output)
-d	Dump packet payload/data
-e	Display link-layer headers (Ethernet, IP, TCP/UDP, ICMP)
-X	Full packet dump in HEX + ASCII
-i	Specify network interface to sniff (e.g., eth0)

1.Sniff on a Specific Interface

```
sudo snort -v -i eth0
```

2.Sniff in Verbose Mode

```
sudo snort -v
```

3.Dump Packet Data (-d)

```
sudo snort -d
```

4. Dump with Link-Layer Headers (-de)

sudo snort -de

5.Full Packet Dump in HEX (-X)

sudo snort -X

10.Operation Mode 2: Packet Logger Mode

Snort can log sniffed packets to files instead of just displaying them. This is called Logger Mode.

Packet Logger Parameters

Parameter	Description
-l	Logger mode — sets the target log directory (default: <code>/var/log/snort</code>)
-K ASCII	Logs packets in human-readable ASCII format
-r	Read mode — read and analyze dumped logs
-n	Process a specific number of packets from a log file

By default, Snort saves logs in binary/tcpdump format, which can be read by Snort, tcpdump, or Wireshark.

1. Logging Packets with -l

```
sudo snort -dev -l .
```

-d → dump packet payload

-e → display link-layer headers

-v → verbose output

-l . → save logs in current directory

2. Logging in ASCII Mode (-K ASCII)

```
Sudo snort -dev -K ASCII -l .
```

3. Reading Logs with -r

```
Sudo snort -r snort.log.5647748
```

4. Opening Logs in tcpdump or Wireshark

```
sudo tcpdump -r snort.log.1638459842 -ntc 10
```

Wireshark can also open binary **.log** files for graphical analysis

11.Operation Mode 3: IDS/IPS

- IDS (Intrusion Detection System):
Monitors network traffic and alerts the user when suspicious activity is detected.
- IPS (Intrusion Prevention System):
Monitors network traffic and blocks malicious packets in real-time.

Both use Snort rules to decide what is “normal” and what should trigger an alert or be blocked.

Example Rule

This simple rule detects **any ICMP packet (ping)** activity in either direction:

```
alert icmp any any <> any any (msg:"ICMP Packet Found"; sid:100001;  
rev:1;)
```

 **Location:** `/etc/snort/rules/local.rules`

When this rule triggers, Snort will create an **alert file**.

Key IDS/IPS Parameters

Parameter	Description
-c	Use a configuration file
-T	Test configuration file

-N	Disable logging
-D	Run in background (daemon) mode
-A	Set alert mode (full, fast, console, cmg, none)

1. Test Configuration

```
sudo snort -c /etc/snort/snort.conf -
```

2. Disable Logging (-N)

```
sudo snort -c /etc/snort/snort.conf -N
```

```
sudo snort -c /etc/snort/snort.conf -N -v(output show on screen No logs saved).
```

3. Background Mode (-D)

```
sudo snort -c /etc/snort/snort.conf -D
```

```
Ps -ef | grep snort
```

```
sudo kill -9 <process_id> to stop
```

4.Alert Modes (-A)

Mode	Description	Output Type
console	Fast alerts on the terminal	On-screen
cmg	Detailed alerts with packet data (hex + text)	On-screen
fast	Minimal alert info (timestamp, IPs, ports)	Log file
full	Detailed alert info	Log file
none	No alerts, only logging	Binary log

-A console Example

```
sudo snort -c /etc/snort/snort.conf -A console
```

Shows real-time alerts like:

```
[**] ICMP Packet Found [**] {ICMP} 192.168.1.10 -> 142.250.187.110
```

-A cmg Example

```
sudo snort -c /etc/snort/snort.conf -A cmg
```

Shows detailed info:

```
[**] ICMP Packet Found [**]
```

```
192.168.1.10 -> 142.250.187.110
```

```
Payload in HEX and text
```

-A fast Example

```
sudo snort -c /etc/snort/snort.conf -A fast
```

Logs minimal alerts to `/var/log/snort/alert`:

```
[**] [1:1000001:0] ICMP Packet Found [**] {ICMP} 192.168.1.10 ->  
8.8.8.8
```

-A full Example

```
sudo snort -c /etc/snort/snort.conf -A full
```

Logs full details of each alert (classification, priority, timestamp, payload, etc.)

-A none Example

```
sudo snort -c /etc/snort/snort.conf -A none
```

No alert file created, but traffic logs are still saved in binary format.

Running Without Configuration File

You can run Snort **with only the rule file**:

```
sudo snort -c /etc/snort/rules/local.rules -A console
```

Useful for testing new or custom rules.

12.Snort in IPS Mode

IPS mode can drop malicious packets instead of just alerting.

Run in IPS Mode:

```
sudo snort -c /etc/snort/snort.conf -Q --daq afpacket -i eth0:eth1 -A console
```

Parameter	Description
-Q	Enable IPS (inline) mode
--daq afpacket	Use DAQ module for packet handling
-i eth0:eth1	Specify input and output interfaces

13.Operation Mode 4: PCAP Investigation

Why Analyze PCAPs with Snort?

When you capture packets with tools like Wireshark or tcpdump, you get *raw traffic data*.

Snort can process these PCAPs using rules to automatically detect suspicious or malicious activity — saving you hours of manual analysis.

Snort PCAP Mode — Key Parameters

Parameter	Description
-r <file>	Read a single PCAP file
--pcap-list="<file1> <file2>"	Read multiple PCAP files at once
--pcap-show	Show PCAP filenames while processing (useful to know which alerts came from which file)
-A console	Display alerts on the console
-n <number>	Limit how many packets to process (useful for testing)
-c <config>	Specify the configuration file (snort.conf)
-q	Quiet mode (hides banners and startup info)

1. Investigate a single PCAP

```
sudo snort -c /etc/snort/snort.conf -q -r icmp-test.pcap -A console -n 10
```

- Use **snort.conf** configuration
 - Quiet mode
 - Read **icmp-test.pcap**
 - Output alerts to console
 - Analyze only 10 packet
-

2. Investigate multiple PCAPs

```
sudo snort -c /etc/snort/snort.conf -q --pcap-list="icmp-test.pcap http2.pcap"  
-A console
```

3. Investigate multiple PCAPs and show filenames

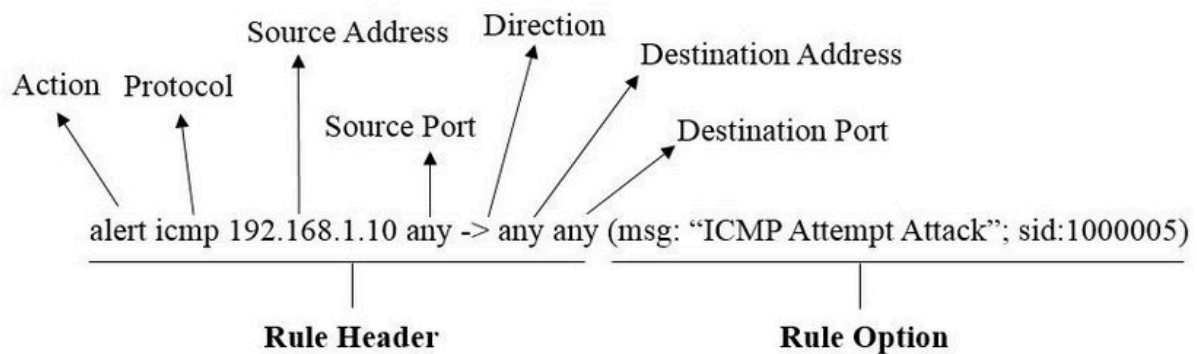
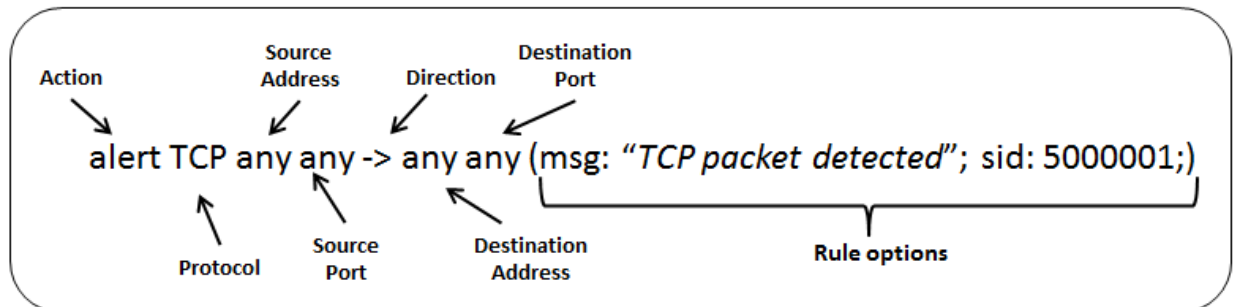
```
sudo snort -c /etc/snort/snort.conf -q --pcap-list="icmp-test.pcap http2.pcap"  
-A console --pcap-show
```

14. What Are Snort Rules?

Snort rules are the *instructions* that tell Snort **what kind of traffic to look for** and **how to respond** when it finds it.

Think of them as “if this happens → do that” conditions for network traffic.

Snort Rule Structure



Every Snort rule has two main parts:

Action Protocol Source_IP Source_Port **Direction** Destination_IP
Destination_Port (Options)

Example

```
alert tcp any any -> any 80 (msg:"HTTP GET Detected"; content:"GET";  
sid:100001; rev:1;)
```

- ◆ **Header** — tells *what traffic* to inspect
- ◆ **Options** — tell *what to do* and *what to look for inside* the packet

Rule Header Breakdown

Part	Meaning	Example
Action	What Snort should do when it matches a rule	alert, log, drop, reject
Protocol	Which protocol to watch	tcp, udp, icmp, ip,http
Source IP/Port	Who is sending the packet	any or 192.168.1.5
Direction	Flow of traffic	-> (one-way) or <> (both ways)
Destination IP/Port	Who is receiving the packet	any, 10.0.0.8, 80, etc.

Common Actions

Action	Meaning
alert	Generate an alert and log the packet (<i>used in IDS mode</i>)
log	Just log the packet
drop	Block and log the packet (<i>used in IPS mode</i>)
reject	Block, log, and send a reset message (<i>IPS mode</i>)

IP and Port Filtering Examples

Use Case	Rule	Meaning
Detect ICMP from 1 IP	alert icmp 192.168.1.56 any <> any any (msg:"ICMP Packet"; sid:100001; rev:1;)	Alerts for every ping from 192.168.1.56
Detect ICMP from subnet	alert icmp 192.168.1.0/24 any <> any any (msg:"ICMP Packet Found"; sid:100002; rev:1;)	Any ping from subnet
Exclude IP range	alert icmp !192.168.1.0/24 any <> any any (msg:"Outside ICMP Packet"; sid:100003; rev:1;)	Alerts on ICMP <i>not</i> from local subnet
Detect FTP	alert tcp any any -> any 21 (msg:"FTP Command Activity"; sid:100004; rev:1;)	Traffic on FTP port 21
Detect System Ports	alert tcp any any -> any 1:1024 (msg:"System Port Activity"; sid:100005; rev:1;)	Detects all TCP traffic to system ports

Direction Operators

Operator	Meaning
->	From source to destination
<>	Both directions
<-	 Doesn't exist in Snort!

Rule Options (inside parentheses)

Rule options provide details about what to look for and what to show in alerts.

Option	Purpose	Example
msg	The message shown when the rule triggers	<code>msg:"ICMP Packet Found";</code>
sid	Unique Snort Rule ID	<code>sid:100001;</code>
rev	Revision number (for updates)	<code>rev:1;</code>
reference	External info like CVE links	<code>reference:cve,CVE-2020-1234;</code>

Payload Rule Options

Option	Purpose	Example
content	Match text or hex in packet payload	<code>content:"GET";</code> or <code>`content:"</code>
nocase	Ignore uppercase/lowercase differences	<code>content:"get"; nocase;</code>
fast_pattern	Speed up searches using a key pattern	<code>content:"GET"; fast_pattern; content:"www";</code>

Non-Payload Options

Option	Purpose	Example
id	Match IP ID field	id:12345;
flags	Match TCP flags (SYN, ACK, etc.)	flags:S;
dsize	Match packet size	dsize:>100;
sameip	Match when source and destination IPs are identical	sameip;

Where to Save Rules

Local rules are stored here:

/etc/snort/rules/local.rules

Edit

Nano /etc/snort/rules/local.rules

Test Your Rule

After saving your rule, test it on a PCAP or live interface:

```
sudo snort -c /etc/snort/snort.conf -A console -l . -r test.pcap
```

-A console → Show alerts in terminal

-l . → Log in current directory

-r → Read from a PCAP file

Main Components of Snort

Component	Description
 Packet Decoder	Collects network packets and prepares them for further analysis.
 Pre-processors	Organize and modify packets (e.g., normalize data) before sending them to the detection engine.
 Detection Engine	The core part of Snort — analyzes packets and checks them against rules to detect threats.
 Logging & Alerting	Records detected events and generates alerts when suspicious activity is found.
 Outputs & Plugins	Handles where the results go (e.g., console, syslog, MySQL) and allows extra plugin functionality.

Types of Snort Rules

Type	Description
Community Rules	Free, open-source rule set (no registration needed).
Registered Rules	Free, but requires registration — contains subscriber rules with a 30-day delay.
Subscriber (Paid) Rules	Paid and updated twice a week — gives the latest and most accurate detections.

To use any of these rule sets, you must include them in your `snort.conf` configuration file.

Important Snort Configuration Files

File Name	Purpose
snort.conf	Main configuration file — manages variables, rules, outputs, and plugins.
local.rules	Custom rules created by the user (your own alerts).

Inside **snort.conf**

You can open it with:

```
sudo gedit /etc/snort/snort.conf
```

Step #1: Set the Network Variables

Defines the scope of protection and rule paths.

Tag Name	Description	Example
HOME_NET	Network you want to protect.	any or 192.168.1.0/24
EXTERNAL_NET	The outside network.	any or !\$HOME_NET
RULE_PATH	Path where rules are stored.	/etc/snort/rules
SO_RULE_PATH	Path for shared object (compiled) rules.	\$RULE_PATH/so_rules
PREPROC_RULE_PATH	Path for preprocessor rules.	\$RULE_PATH/plugin_rules

Step #2: Configure the Decoder

Manages how Snort captures and inspects packets (IPS or IDS mode).

Tag Name	Description	Example
config daq:	Selects DAQ (Data Acquisition) module type.	afpacket
config daq_mode:	Chooses operation mode (inline for IPS).	inline
config logdir:	Defines where Snort stores logs.	/var/logs/snort

DAQ (Data Acquisition) Modules

Module	Type	Description
pcap	Sniffer mode	Default — passive packet capture.
afpacket	Inline mode	Best for IPS (real-time protection).
ipq	Inline (Linux)	Uses Netfilter; replaces snort_inline patch.
nfq	Inline (Linux)	Uses NFQUEUE for packet handling.
ipfw	Inline (BSD)	For FreeBSD/OpenBSD firewalls.
dump	Testing	Used for testing and debugging.

Commonly used:

- `pcap` → for IDS mode
 - `afpacket` → for IPS (inline) mode
-

Step #6: Configure Output Plugins

Defines how and where Snort sends alerts/logs.

Example:

- To console (default)
- To files, syslog, or databases
- To third-party tools like SIEMs

Customizing output plugins helps in **better visibility** and **integration** with monitoring tools.

Step #7: Customise Your Ruleset

Defines which rules Snort will use.

Tag Name	Description	Example
# site specific rules	Local user-created rules.	include \$RULE_PATH/local.rules
# include \$RULE_PATH/	Default/downloaded rule sets.	include \$RULE_PATH/rulename

Note:

Lines starting with # are comments.

Remove # to activate a line (uncomment).

Reference:

Download - <https://www.snort.org/>

Rules Documentation- <https://docs.snort.org/welcome>

User manual -<http://manual-snort-org.s3-website-us-east-1.amazonaws.com/>
[https://en.wikipedia.org/wiki/Snort_\(software\)](https://en.wikipedia.org/wiki/Snort_(software))

-yuvaraj.D

